SG No. 213 O^7 $P4_132$ [cubic]

* plus set: $+[0,\ 0,\ 0]$
* Wyckoff site: `4a`, site symmetry: `.32`

Table 1: Wyckoff bond: `4a@4a`, bond symmetry: `.32`

No.	vector	center	mapping
1	$[X,\ X,\ X]$	$[\frac{3}{8},\ \frac{3}{8},\ \frac{3}{8}]$	`[1,5,9,-14,-19,-24]`
2	$[-X,\ -X,\ X]$	$[\frac{1}{8},\ \frac{5}{8},\ \frac{7}{8}]$	`[2,7,12,-13,-17,-21]`
3	$[-X,\ X,\ -X]$	$[\frac{5}{8},\ \frac{7}{8},\ \frac{1}{8}]$	`[3,8,10,-15,-20,-22]`
4	$[X,\ -X,\ -X]$	$[\frac{7}{8},\ \frac{1}{8},\ \frac{5}{8}]$	`[4,6,11,-16,-18,-23]`

Table 2: Wyckoff bond: `12b@4a`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\ X,\ Y]$	$[\frac{3}{8},\ \frac{3}{8},\ \frac{3}{8}]$	`[1,-14]`
2	$[-X,\ -X,\ Y]$	$[\frac{1}{8},\ \frac{5}{8},\ \frac{7}{8}]$	`[2,-13]`
3	$[-X,\ X,\ -Y]$	$[\frac{5}{8},\ \frac{7}{8},\ \frac{1}{8}]$	`[3,-15]`
4	$[X,\ -X,\ -Y]$	$[\frac{7}{8},\ \frac{1}{8},\ \frac{5}{8}]$	`[4,-16]`
5	$[Y,\ X,\ X]$	$[\frac{3}{8},\ \frac{3}{8},\ \frac{3}{8}]$	`[5,-24]`
6	$[Y,\ -X,\ -X]$	$[\frac{7}{8},\ \frac{1}{8},\ \frac{5}{8}]$	`[6,-23]`
7	$[-Y,\ -X,\ X]$	$[\frac{1}{8},\ \frac{5}{8},\ \frac{7}{8}]$	`[7,-21]`
8	$[-Y,\ X,\ -X]$	$[\frac{5}{8},\ \frac{7}{8},\ \frac{1}{8}]$	`[8,-22]`
9	$[X,\ Y,\ X]$	$[\frac{3}{8},\ \frac{3}{8},\ \frac{3}{8}]$	`[9,-19]`
10	$[-X,\ Y,\ -X]$	$[\frac{5}{8},\ \frac{7}{8},\ \frac{1}{8}]$	`[10,-20]`
11	$[X,\ -Y,\ -X]$	$[\frac{7}{8},\ \frac{1}{8},\ \frac{5}{8}]$	`[11,-18]`
12	$[-X,\ -Y,\ X]$	$[\frac{1}{8},\ \frac{5}{8},\ \frac{7}{8}]$	`[12,-17]`

Table 3: Wyckoff bond: `12c@4a`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\ -X,\ 0]$	$[\frac{3}{8},\ \frac{3}{8},\ \frac{3}{8}]$	`[1,14]`
2	$[-X,\ X,\ 0]$	$[\frac{1}{8},\ \frac{5}{8},\ \frac{7}{8}]$	`[2,13]`
3	$[-X,\ -X,\ 0]$	$[\frac{5}{8},\ \frac{7}{8},\ \frac{1}{8}]$	`[3,15]`
4	$[X,\ X,\ 0]$	$[\frac{7}{8},\ \frac{1}{8},\ \frac{5}{8}]$	`[4,16]`
5	$[0,\ X,\ -X]$	$[\frac{3}{8},\ \frac{3}{8},\ \frac{3}{8}]$	`[5,24]`
6	$[0,\ -X,\ X]$	$[\frac{7}{8},\ \frac{1}{8},\ \frac{5}{8}]$	`[6,23]`
7	$[0,\ -X,\ -X]$	$[\frac{1}{8},\ \frac{5}{8},\ \frac{7}{8}]$	`[7,21]`
8	$[0,\ X,\ X]$	$[\frac{5}{8},\ \frac{7}{8},\ \frac{1}{8}]$	`[8,22]`
9	$[-X,\ 0,\ X]$	$[\frac{3}{8},\ \frac{3}{8},\ \frac{3}{8}]$	`[9,19]`
10	$[X,\ 0,\ -X]$	$[\frac{5}{8},\ \frac{7}{8},\ \frac{1}{8}]$	`[10,20]`
11	$[-X,\ 0,\ -X]$	$[\frac{7}{8},\ \frac{1}{8},\ \frac{5}{8}]$	`[11,18]`
12	$[X,\ 0,\ X]$	$[\frac{1}{8},\ \frac{5}{8},\ \frac{7}{8}]$	`[12,17]`

Table 4: Wyckoff bond: `24d@4a`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[1]`
2	$[-X, -Y, Z]$	$[\frac{1}{8}, \frac{5}{8}, \frac{7}{8}]$	`[2]`
3	$[-X, Y, -Z]$	$[\frac{5}{8}, \frac{7}{8}, \frac{1}{8}]$	`[3]`
4	$[X, -Y, -Z]$	$[\frac{7}{8}, \frac{1}{8}, \frac{5}{8}]$	`[4]`
5	$[Z, X, Y]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[5]`
6	$[Z, -X, -Y]$	$[\frac{7}{8}, \frac{1}{8}, \frac{5}{8}]$	`[6]`
7	$[-Z, -X, Y]$	$[\frac{1}{8}, \frac{5}{8}, \frac{7}{8}]$	`[7]`
8	$[-Z, X, -Y]$	$[\frac{5}{8}, \frac{7}{8}, \frac{1}{8}]$	`[8]`
9	$[Y, Z, X]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[9]`
10	$[-Y, Z, -X]$	$[\frac{5}{8}, \frac{7}{8}, \frac{1}{8}]$	`[10]`
11	$[Y, -Z, -X]$	$[\frac{7}{8}, \frac{1}{8}, \frac{5}{8}]$	`[11]`
12	$[-Y, -Z, X]$	$[\frac{1}{8}, \frac{5}{8}, \frac{7}{8}]$	`[12]`
13	$[Y, X, -Z]$	$[\frac{1}{8}, \frac{5}{8}, \frac{7}{8}]$	`[13]`
14	$[-Y, -X, -Z]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[14]`
15	$[Y, -X, Z]$	$[\frac{5}{8}, \frac{7}{8}, \frac{1}{8}]$	`[15]`
16	$[-Y, X, Z]$	$[\frac{7}{8}, \frac{1}{8}, \frac{5}{8}]$	`[16]`
17	$[X, Z, -Y]$	$[\frac{1}{8}, \frac{5}{8}, \frac{7}{8}]$	`[17]`
18	$[-X, Z, Y]$	$[\frac{7}{8}, \frac{1}{8}, \frac{5}{8}]$	`[18]`
19	$[-X, -Z, -Y]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[19]`
20	$[X, -Z, Y]$	$[\frac{5}{8}, \frac{7}{8}, \frac{1}{8}]$	`[20]`
21	$[Z, Y, -X]$	$[\frac{1}{8}, \frac{5}{8}, \frac{7}{8}]$	`[21]`
22	$[Z, -Y, X]$	$[\frac{5}{8}, \frac{7}{8}, \frac{1}{8}]$	`[22]`
23	$[-Z, Y, X]$	$[\frac{7}{8}, \frac{1}{8}, \frac{5}{8}]$	`[23]`
24	$[-Z, -Y, -X]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[24]`

* Wyckoff site: `4b`, site symmetry: `.32`

Table 5: Wyckoff bond: `4a@4b`, bond symmetry: `.32`

No.	vector	center	mapping
1	$[X, X, X]$	$[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}]$	`[1,5,9,-14,-19,-24]`
2	$[-X, -X, X]$	$[\frac{5}{8}, \frac{1}{8}, \frac{3}{8}]$	`[2,7,12,-13,-17,-21]`
3	$[-X, X, -X]$	$[\frac{1}{8}, \frac{3}{8}, \frac{5}{8}]$	`[3,8,10,-15,-20,-22]`
4	$[X, -X, -X]$	$[\frac{3}{8}, \frac{5}{8}, \frac{1}{8}]$	`[4,6,11,-16,-18,-23]`

Table 6: Wyckoff bond: `12b@4b`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, X, Y]$	$[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}]$	`[1,-14]`
2	$[-X, -X, Y]$	$[\frac{5}{8}, \frac{1}{8}, \frac{3}{8}]$	`[2,-13]`
3	$[-X, X, -Y]$	$[\frac{1}{8}, \frac{3}{8}, \frac{5}{8}]$	`[3,-15]`
4	$[X, -X, -Y]$	$[\frac{3}{8}, \frac{5}{8}, \frac{1}{8}]$	`[4,-16]`
5	$[Y, X, X]$	$[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}]$	`[5,-24]`
6	$[Y, -X, -X]$	$[\frac{3}{8}, \frac{5}{8}, \frac{1}{8}]$	`[6,-23]`
7	$[-Y, -X, X]$	$[\frac{5}{8}, \frac{1}{8}, \frac{3}{8}]$	`[7,-21]`

continued ...

Table 6

No.	vector	center	mapping
8	$[-Y, X, -X]$	$[\frac{1}{8}, \frac{3}{8}, \frac{5}{8}]$	`[8,-22]`
9	$[X, Y, X]$	$[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}]$	`[9,-19]`
10	$[-X, Y, -X]$	$[\frac{1}{8}, \frac{3}{8}, \frac{5}{8}]$	`[10,-20]`
11	$[X, -Y, -X]$	$[\frac{3}{8}, \frac{5}{8}, \frac{1}{8}]$	`[11,-18]`
12	$[-X, -Y, X]$	$[\frac{5}{8}, \frac{1}{8}, \frac{3}{8}]$	`[12,-17]`

Table 7: Wyckoff bond: `12c@4b`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, -X, 0]$	$[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}]$	`[1,14]`
2	$[-X, X, 0]$	$[\frac{5}{8}, \frac{1}{8}, \frac{3}{8}]$	`[2,13]`
3	$[-X, -X, 0]$	$[\frac{1}{8}, \frac{3}{8}, \frac{5}{8}]$	`[3,15]`
4	$[X, X, 0]$	$[\frac{3}{8}, \frac{5}{8}, \frac{1}{8}]$	`[4,16]`
5	$[0, X, -X]$	$[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}]$	`[5,24]`
6	$[0, -X, X]$	$[\frac{3}{8}, \frac{5}{8}, \frac{1}{8}]$	`[6,23]`
7	$[0, -X, -X]$	$[\frac{5}{8}, \frac{1}{8}, \frac{3}{8}]$	`[7,21]`
8	$[0, X, X]$	$[\frac{1}{8}, \frac{3}{8}, \frac{5}{8}]$	`[8,22]`
9	$[-X, 0, X]$	$[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}]$	`[9,19]`
10	$[X, 0, -X]$	$[\frac{1}{8}, \frac{3}{8}, \frac{5}{8}]$	`[10,20]`
11	$[-X, 0, -X]$	$[\frac{3}{8}, \frac{5}{8}, \frac{1}{8}]$	`[11,18]`
12	$[X, 0, X]$	$[\frac{5}{8}, \frac{1}{8}, \frac{3}{8}]$	`[12,17]`

Table 8: Wyckoff bond: `24d@4b`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}]$	`[1]`
2	$[-X, -Y, Z]$	$[\frac{5}{8}, \frac{1}{8}, \frac{3}{8}]$	`[2]`
3	$[-X, Y, -Z]$	$[\frac{1}{8}, \frac{3}{8}, \frac{5}{8}]$	`[3]`
4	$[X, -Y, -Z]$	$[\frac{3}{8}, \frac{5}{8}, \frac{1}{8}]$	`[4]`
5	$[Z, X, Y]$	$[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}]$	`[5]`
6	$[Z, -X, -Y]$	$[\frac{3}{8}, \frac{5}{8}, \frac{1}{8}]$	`[6]`
7	$[-Z, -X, Y]$	$[\frac{5}{8}, \frac{1}{8}, \frac{3}{8}]$	`[7]`
8	$[-Z, X, -Y]$	$[\frac{1}{8}, \frac{3}{8}, \frac{5}{8}]$	`[8]`
9	$[Y, Z, X]$	$[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}]$	`[9]`
10	$[-Y, Z, -X]$	$[\frac{1}{8}, \frac{3}{8}, \frac{5}{8}]$	`[10]`
11	$[Y, -Z, -X]$	$[\frac{3}{8}, \frac{5}{8}, \frac{1}{8}]$	`[11]`
12	$[-Y, -Z, X]$	$[\frac{5}{8}, \frac{1}{8}, \frac{3}{8}]$	`[12]`
13	$[Y, X, -Z]$	$[\frac{5}{8}, \frac{1}{8}, \frac{3}{8}]$	`[13]`
14	$[-Y, -X, -Z]$	$[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}]$	`[14]`
15	$[Y, -X, Z]$	$[\frac{1}{8}, \frac{3}{8}, \frac{5}{8}]$	`[15]`
16	$[-Y, X, Z]$	$[\frac{3}{8}, \frac{5}{8}, \frac{1}{8}]$	`[16]`
17	$[X, Z, -Y]$	$[\frac{5}{8}, \frac{1}{8}, \frac{3}{8}]$	`[17]`
18	$[-X, Z, Y]$	$[\frac{3}{8}, \frac{5}{8}, \frac{1}{8}]$	`[18]`
19	$[-X, -Z, -Y]$	$[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}]$	`[19]`

continued ...

Table 8

No.	vector	center	mapping
20	$[X, -Z, Y]$	$\left[\frac{1}{8}, \frac{3}{8}, \frac{5}{8}\right]$	[20]
21	$[Z, Y, -X]$	$\left[\frac{5}{8}, \frac{1}{8}, \frac{3}{8}\right]$	[21]
22	$[Z, -Y, X]$	$\left[\frac{1}{8}, \frac{3}{8}, \frac{5}{8}\right]$	[22]
23	$[-Z, Y, X]$	$\left[\frac{3}{8}, \frac{5}{8}, \frac{1}{8}\right]$	[23]
24	$[-Z, -Y, -X]$	$\left[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}\right]$	[24]

* Wyckoff site: 8c, site symmetry: .3.

Table 9: Wyckoff bond: 8a@8c, bond symmetry: .3.

No.	vector	center	mapping
1	$[X, X, X]$	$[x, x, x]$	[1,5,9]
2	$[-X, -X, X]$	$\left[\frac{1}{2} - x, -x, x + \frac{1}{2}\right]$	[2,7,12]
3	$[-X, X, -X]$	$\left[-x, x + \frac{1}{2}, \frac{1}{2} - x\right]$	[3,8,10]
4	$[X, -X, -X]$	$\left[x + \frac{1}{2}, \frac{1}{2} - x, -x\right]$	[4,6,11]
5	$[X, X, -X]$	$\left[x + \frac{3}{4}, x + \frac{1}{4}, \frac{1}{4} - x\right]$	[13,17,21]
6	$[-X, -X, -X]$	$\left[\frac{3}{4} - x, \frac{3}{4} - x, \frac{3}{4} - x\right]$	[14,19,24]
7	$[X, -X, X]$	$\left[x + \frac{1}{4}, \frac{1}{4} - x, x + \frac{3}{4}\right]$	[15,20,22]
8	$[-X, X, X]$	$\left[\frac{1}{4} - x, x + \frac{3}{4}, x + \frac{1}{4}\right]$	[16,18,23]

Table 10: Wyckoff bond: 24b@8c, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, x, x]$	[1]
2	$[-X, -Y, Z]$	$\left[\frac{1}{2} - x, -x, x + \frac{1}{2}\right]$	[2]
3	$[-X, Y, -Z]$	$\left[-x, x + \frac{1}{2}, \frac{1}{2} - x\right]$	[3]
4	$[X, -Y, -Z]$	$\left[x + \frac{1}{2}, \frac{1}{2} - x, -x\right]$	[4]
5	$[Z, X, Y]$	$[x, x, x]$	[5]
6	$[Z, -X, -Y]$	$\left[x + \frac{1}{2}, \frac{1}{2} - x, -x\right]$	[6]
7	$[-Z, -X, Y]$	$\left[\frac{1}{2} - x, -x, x + \frac{1}{2}\right]$	[7]
8	$[-Z, X, -Y]$	$\left[-x, x + \frac{1}{2}, \frac{1}{2} - x\right]$	[8]
9	$[Y, Z, X]$	$[x, x, x]$	[9]
10	$[-Y, Z, -X]$	$\left[-x, x + \frac{1}{2}, \frac{1}{2} - x\right]$	[10]
11	$[Y, -Z, -X]$	$\left[x + \frac{1}{2}, \frac{1}{2} - x, -x\right]$	[11]
12	$[-Y, -Z, X]$	$\left[\frac{1}{2} - x, -x, x + \frac{1}{2}\right]$	[12]
13	$[Y, X, -Z]$	$\left[x + \frac{3}{4}, x + \frac{1}{4}, \frac{1}{4} - x\right]$	[13]
14	$[-Y, -X, -Z]$	$\left[\frac{3}{4} - x, \frac{3}{4} - x, \frac{3}{4} - x\right]$	[14]
15	$[Y, -X, Z]$	$\left[x + \frac{1}{4}, \frac{1}{4} - x, x + \frac{3}{4}\right]$	[15]
16	$[-Y, X, Z]$	$\left[\frac{1}{4} - x, x + \frac{3}{4}, x + \frac{1}{4}\right]$	[16]
17	$[X, Z, -Y]$	$\left[x + \frac{3}{4}, x + \frac{1}{4}, \frac{1}{4} - x\right]$	[17]
18	$[-X, Z, Y]$	$\left[\frac{1}{4} - x, x + \frac{3}{4}, x + \frac{1}{4}\right]$	[18]
19	$[-X, -Z, -Y]$	$\left[\frac{3}{4} - x, \frac{3}{4} - x, \frac{3}{4} - x\right]$	[19]
20	$[X, -Z, Y]$	$\left[x + \frac{1}{4}, \frac{1}{4} - x, x + \frac{3}{4}\right]$	[20]
21	$[Z, Y, -X]$	$\left[x + \frac{3}{4}, x + \frac{1}{4}, \frac{1}{4} - x\right]$	[21]
22	$[Z, -Y, X]$	$\left[x + \frac{1}{4}, \frac{1}{4} - x, x + \frac{3}{4}\right]$	[22]
23	$[-Z, Y, X]$	$\left[\frac{1}{4} - x, x + \frac{3}{4}, x + \frac{1}{4}\right]$	[23]

continued ...

Table 10

No.	vector	center	mapping
24	$[-Z, -Y, -X]$	$\left[\frac{3}{4} - x, \frac{3}{4} - x, \frac{3}{4} - x\right]$	[24]

* Wyckoff site: 12d, site symmetry: ..2

Table 11: Wyckoff bond: 12a@12d, bond symmetry: ..2

No.	vector	center	mapping
1	$[Y, X, -X]$	$\left[\frac{1}{8}, y, y + \frac{1}{4}\right]$	[1,-18]
2	$[-Y, -X, -X]$	$\left[\frac{3}{8}, -y, y + \frac{3}{4}\right]$	[2,-20]
3	$[-Y, X, X]$	$\left[\frac{7}{8}, y + \frac{1}{2}, \frac{1}{4} - y\right]$	[3,-17]
4	$[Y, -X, X]$	$\left[\frac{5}{8}, \frac{1}{2} - y, \frac{3}{4} - y\right]$	[4,-19]
5	$[-X, Y, X]$	$\left[y + \frac{1}{4}, \frac{1}{8}, y\right]$	[5,-15]
6	$[-X, -Y, -X]$	$\left[y + \frac{3}{4}, \frac{3}{8}, -y\right]$	[6,-13]
7	$[X, -Y, X]$	$\left[\frac{1}{4} - y, \frac{7}{8}, y + \frac{1}{2}\right]$	[7,-16]
8	$[X, Y, -X]$	$\left[\frac{3}{4} - y, \frac{5}{8}, \frac{1}{2} - y\right]$	[8,-14]
9	$[X, -X, Y]$	$\left[y, y + \frac{1}{4}, \frac{1}{8}\right]$	[9,-21]
10	$[-X, -X, -Y]$	$\left[-y, y + \frac{3}{4}, \frac{3}{8}\right]$	[10,-23]
11	$[X, X, -Y]$	$\left[y + \frac{1}{2}, \frac{1}{4} - y, \frac{7}{8}\right]$	[11,-22]
12	$[-X, X, Y]$	$\left[\frac{1}{2} - y, \frac{3}{4} - y, \frac{5}{8}\right]$	[12,-24]

Table 12: Wyckoff bond: 12b@12d, bond symmetry: ..2

No.	vector	center	mapping
1	$[0, X, X]$	$\left[\frac{1}{8}, y, y + \frac{1}{4}\right]$	[1,18]
2	$[0, -X, X]$	$\left[\frac{3}{8}, -y, y + \frac{3}{4}\right]$	[2,20]
3	$[0, X, -X]$	$\left[\frac{7}{8}, y + \frac{1}{2}, \frac{1}{4} - y\right]$	[3,17]
4	$[0, -X, -X]$	$\left[\frac{5}{8}, \frac{1}{2} - y, \frac{3}{4} - y\right]$	[4,19]
5	$[X, 0, X]$	$\left[y + \frac{1}{4}, \frac{1}{8}, y\right]$	[5,15]
6	$[X, 0, -X]$	$\left[y + \frac{3}{4}, \frac{3}{8}, -y\right]$	[6,13]
7	$[-X, 0, X]$	$\left[\frac{1}{4} - y, \frac{7}{8}, y + \frac{1}{2}\right]$	[7,16]
8	$[-X, 0, -X]$	$\left[\frac{3}{4} - y, \frac{5}{8}, \frac{1}{2} - y\right]$	[8,14]
9	$[X, X, 0]$	$\left[y, y + \frac{1}{4}, \frac{1}{8}\right]$	[9,21]
10	$[-X, X, 0]$	$\left[-y, y + \frac{3}{4}, \frac{3}{8}\right]$	[10,23]
11	$[X, -X, 0]$	$\left[y + \frac{1}{2}, \frac{1}{4} - y, \frac{7}{8}\right]$	[11,22]
12	$[-X, -X, 0]$	$\left[\frac{1}{2} - y, \frac{3}{4} - y, \frac{5}{8}\right]$	[12,24]

Table 13: Wyckoff bond: 24c@12d, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{8}, y, y + \frac{1}{4}\right]$	[1]
2	$[-X, -Y, Z]$	$\left[\frac{3}{8}, -y, y + \frac{3}{4}\right]$	[2]
3	$[-X, Y, -Z]$	$\left[\frac{7}{8}, y + \frac{1}{2}, \frac{1}{4} - y\right]$	[3]
4	$[X, -Y, -Z]$	$\left[\frac{5}{8}, \frac{1}{2} - y, \frac{3}{4} - y\right]$	[4]
5	$[Z, X, Y]$	$\left[y + \frac{1}{4}, \frac{1}{8}, y\right]$	[5]

continued ...

Table 13

No.	vector	center	mapping
6	$[Z, -X, -Y]$	$\left[y+\frac{3}{4}, \frac{3}{8}, -y\right]$	[6]
7	$[-Z, -X, Y]$	$\left[\frac{1}{4}-y, \frac{7}{8}, y+\frac{1}{2}\right]$	[7]
8	$[-Z, X, -Y]$	$\left[\frac{3}{4}-y, \frac{5}{8}, \frac{1}{2}-y\right]$	[8]
9	$[Y, Z, X]$	$\left[y, y+\frac{1}{4}, \frac{1}{8}\right]$	[9]
10	$[-Y, Z, -X]$	$\left[-y, y+\frac{3}{4}, \frac{3}{8}\right]$	[10]
11	$[Y, -Z, -X]$	$\left[y+\frac{1}{2}, \frac{1}{4}-y, \frac{7}{8}\right]$	[11]
12	$[-Y, -Z, X]$	$\left[\frac{1}{2}-y, \frac{3}{4}-y, \frac{5}{8}\right]$	[12]
13	$[Y, X, -Z]$	$\left[y+\frac{3}{4}, \frac{3}{8}, -y\right]$	[13]
14	$[-Y, -X, -Z]$	$\left[\frac{3}{4}-y, \frac{5}{8}, \frac{1}{2}-y\right]$	[14]
15	$[Y, -X, Z]$	$\left[y+\frac{1}{4}, \frac{1}{8}, y\right]$	[15]
16	$[-Y, X, Z]$	$\left[\frac{1}{4}-y, \frac{7}{8}, y+\frac{1}{2}\right]$	[16]
17	$[X, Z, -Y]$	$\left[\frac{7}{8}, y+\frac{1}{2}, \frac{1}{4}-y\right]$	[17]
18	$[-X, Z, Y]$	$\left[\frac{1}{8}, y, y+\frac{1}{4}\right]$	[18]
19	$[-X, -Z, -Y]$	$\left[\frac{5}{8}, \frac{1}{2}-y, \frac{3}{4}-y\right]$	[19]
20	$[X, -Z, Y]$	$\left[\frac{3}{8}, -y, y+\frac{3}{4}\right]$	[20]
21	$[Z, Y, -X]$	$\left[y, y+\frac{1}{4}, \frac{1}{8}\right]$	[21]
22	$[Z, -Y, X]$	$\left[y+\frac{1}{2}, \frac{1}{4}-y, \frac{7}{8}\right]$	[22]
23	$[-Z, Y, X]$	$\left[-y, y+\frac{3}{4}, \frac{3}{8}\right]$	[23]
24	$[-Z, -Y, -X]$	$\left[\frac{1}{2}-y, \frac{3}{4}-y, \frac{5}{8}\right]$	[24]

* Wyckoff site: 24e, site symmetry: 1

Table 14: Wyckoff bond: 24a@24e, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, z]$	[1]
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}-x, -y, z+\frac{1}{2}\right]$	[2]
3	$[-X, Y, -Z]$	$\left[-x, y+\frac{1}{2}, \frac{1}{2}-z\right]$	[3]
4	$[X, -Y, -Z]$	$\left[x+\frac{1}{2}, \frac{1}{2}-y, -z\right]$	[4]
5	$[Z, X, Y]$	$[z, x, y]$	[5]
6	$[Z, -X, -Y]$	$\left[z+\frac{1}{2}, \frac{1}{2}-x, -y\right]$	[6]
7	$[-Z, -X, Y]$	$\left[\frac{1}{2}-z, -x, y+\frac{1}{2}\right]$	[7]
8	$[-Z, X, -Y]$	$\left[-z, x+\frac{1}{2}, \frac{1}{2}-y\right]$	[8]
9	$[Y, Z, X]$	$[y, z, x]$	[9]
10	$[-Y, Z, -X]$	$\left[-y, z+\frac{1}{2}, \frac{1}{2}-x\right]$	[10]
11	$[Y, -Z, -X]$	$\left[y+\frac{1}{2}, \frac{1}{2}-z, -x\right]$	[11]
12	$[-Y, -Z, X]$	$\left[\frac{1}{2}-y, -z, x+\frac{1}{2}\right]$	[12]
13	$[Y, X, -Z]$	$\left[y+\frac{3}{4}, x+\frac{1}{4}, \frac{1}{4}-z\right]$	[13]
14	$[-Y, -X, -Z]$	$\left[\frac{3}{4}-y, \frac{3}{4}-x, \frac{3}{4}-z\right]$	[14]
15	$[Y, -X, Z]$	$\left[y+\frac{1}{4}, \frac{1}{4}-x, z+\frac{3}{4}\right]$	[15]
16	$[-Y, X, Z]$	$\left[\frac{1}{4}-y, x+\frac{3}{4}, z+\frac{1}{4}\right]$	[16]
17	$[X, Z, -Y]$	$\left[x+\frac{3}{4}, z+\frac{1}{4}, \frac{1}{4}-y\right]$	[17]
18	$[-X, Z, Y]$	$\left[\frac{1}{4}-x, z+\frac{3}{4}, y+\frac{1}{4}\right]$	[18]
19	$[-X, -Z, -Y]$	$\left[\frac{3}{4}-x, \frac{3}{4}-z, \frac{3}{4}-y\right]$	[19]
20	$[X, -Z, Y]$	$\left[x+\frac{1}{4}, \frac{1}{4}-z, y+\frac{3}{4}\right]$	[20]
21	$[Z, Y, -X]$	$\left[z+\frac{3}{4}, y+\frac{1}{4}, \frac{1}{4}-x\right]$	[21]
22	$[Z, -Y, X]$	$\left[z+\frac{1}{4}, \frac{1}{4}-y, x+\frac{3}{4}\right]$	[22]

continued ...

Table 14

No.	vector	center	mapping
23	$[-Z,\, Y,\, X]$	$\left[\frac{1}{4} - z,\, y + \frac{3}{4},\, x + \frac{1}{4}\right]$	[23]
24	$[-Z,\, -Y,\, -X]$	$\left[\frac{3}{4} - z,\, \frac{3}{4} - y,\, \frac{3}{4} - x\right]$	[24]